

BRAINBYTE

Personal AI Tutor



Internship Group Project

Hindu College ,Amritsar

BCA 3rd Semester



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Roll No:- 25267513

Member 2 – AI Tutor & Personalization

02 | Problem Statement

- Every student have a different learning level.
- Student may be strong in some concepts But weak in others.
- Traditional learning may provide the same Content to every student.
- General AI tools may answer questions but do not automatically track the student's learning journey.
- Student need personalized guidance based on their performance



**The main problem is that every student learns differently,
But traditional learning may not always personalize the learning process.**

03 | Our Solution - BrainByte



BrainByte is an AI-powered Personal Tutor that adapts lesson and practice according to a student learning level and performances.



Student Profile
(name, age, class, goal)



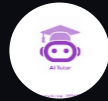
Adaptive Quiz
(based on performance)



Personalized AI Lesson
(Simple + easy explanation)



Progress Tracking
(shows improvement)



AI Tutor
(Personalized guide based on student performance)



Weak topic Analysis
(focus on what you need)

Student Profile Page

TECH STACK

Technologies used to build the Personal AI Tutor

Python
Core Programming Language

Streamlit
Interactive web Application

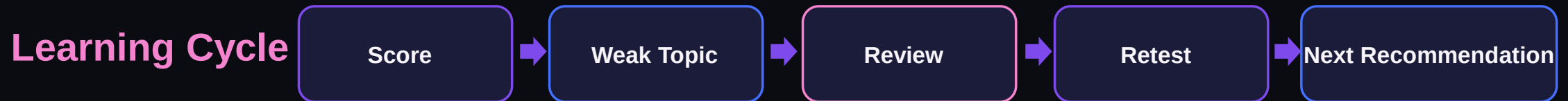
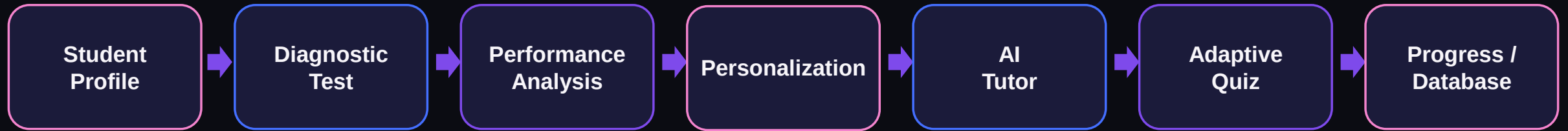
OpenAI API
AI-powered personalized tutoring

Pandas
Data handling and performance analysis.

SQLite
Student and performance data storage.

Git & GitHub
Version control & collaboration

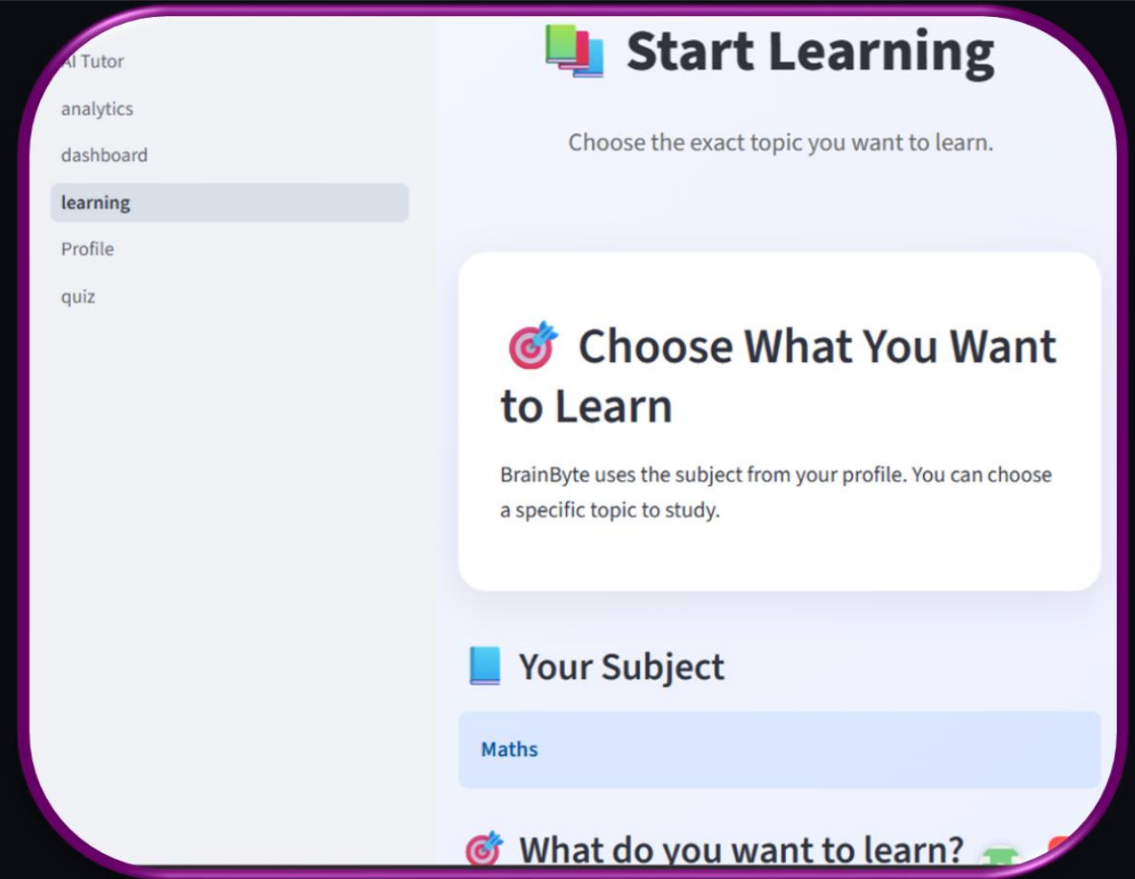
05 | System Architecture



06 | Live Demo

What the demo shows

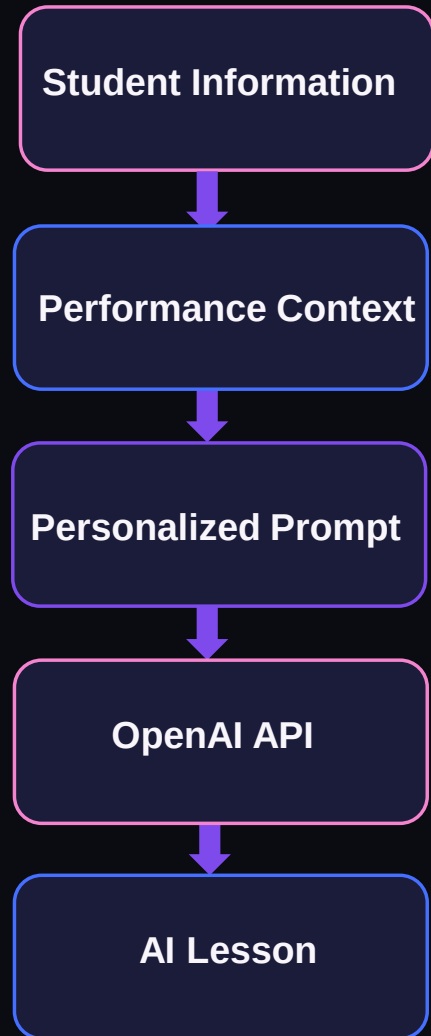
- **Student profile** – Enter student details and learning goal.
- **Choose a Topic** - Select a subject and topic to learn.
- **Personalized AI Lessons** – AI generates an explanation based on the student needs.
- **Quiz & Feedback** – Student practices, gets a score and identifies weak areas.



Project link:-

Student Profile Page

<https://ai-tutor-ei2dq9mh7qlq4eekux2ngf.streamlit.app/>



What the AI Tutor does

1. Uses the student's subject, topic and learning level.
2. Include performance and weak concepts in the AI Prompt.
3. OpenAI API generates a lesson based on the student needs.
4. Provide simple explanations and worked examples.

08 | Challenges & Solutions

Personalized AI Prompt

Challenge: Generic prompts can produce generic responses.

Solution: Include topic, level, goals and weak concepts in the prompt.

API / AI Failures

Challenge: An API issue can interrupt the tutor experience.

Solution: Use error handling and show a useful user message.

Connecting Modules

Challenge: Profile, performance, AI lesson and quiz must work together.

Solution: Pass relevant student data through a structured workflow.

Simple User Experience

Challenge: Students should not feel lost in the learning flow.

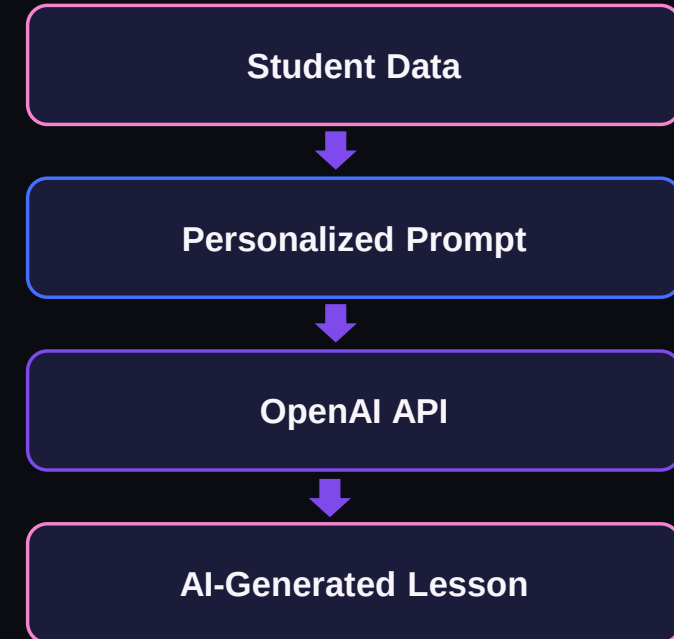
Solution: Use clear sections, labels and step-by-step navigation.

MEMBER 2 – AI TUTOR & PERSONALIZATION

MANVEET KAUR • ROLL NO. 25267513

- Designed the personalized AI Tutor flow.
- Created prompts based on the student's topic and learning level.
- Integrated the OpenAI API for AI-generated learning content.
- Used student performance and weak concepts for personalization.
- Worked on the AI Tutor interface and learning workflow.

MY AI FLOW



Focus: AI Integration + Personalized Learning

10 | Future Improvements

Voice-Based AI Tutor

Add voice interaction so students can learn by speaking and listening.

More Subjects & Topics

Expand BrainByte to support more subjects and learning topics.

Advanced Progress Analytics

Provide detailed insights about scores, weak areas and learning progress.

Smarter Learning Recommendations

Recommend the next topic, quiz or revision activity according to performance.

Goal: Make learning more personalized, accessible and data-driven.

PROJECT REPOSITORY

BrainByte – Source Code & Project Files

<https://github.com/Sneha4-web/AI-Tutor.git>

SOURCE CODE

Project code and development files

DOCUMENTATION

README and supporting project material

COLLABORATION

Git & GitHub for version control

THANK YOU

BrainByte – Personal AI Tutor

Questions & Answers

Manveet Kaur • Sneha • Priya

BCA – 3rd Semester | Hindu College

